RenoFusion: Diagnosing the Retrieval–Fusion Calibration Hazard in Decision-Level Multimodal Fusion and LLM-Powered Retrieval-Augmented Generation for Metastasis Prediction in Clear Cell Renal Cell Carcinoma

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ABSTRACT

Multimodal decision-level fusion and retrieval-augmented generation (RAG) are increasingly adopted for clinical decision support in oncology, yet neither paradigm systematically evaluates how uncalibrated score distributions and class-imbalance rebalancing artifacts distort downstream inference. We introduce the Retrieval–Fusion Calibration Hazard, an evaluation artifact wherein ad-hoc rank normalization of heterogeneous retrieval scores (unbounded BM25, bounded dense cosine, and bounded radiomic similarity) inflates apparent retrieval and generation metrics, directly paralleling log-odds distortion in classifier fusion. This study presents RenoFusion, a two-stage open-source framework evaluated on an aligned cohort of $n = 126$ TCGA-KIRC patients (18 M_1; 14.29% prevalence). In Stage I (3D Multimodal Decision Fusion), three single-modality base models (clinical TNM staging, ssGSEA hallmark pathway genomics, and 3D volumetric CT deep radiomics combining PyRadiomics with MedicalNet ResNet) and seven decision-level fusion architectures were benchmarked across calibration regimes using 2,000 paired bootstrap resamples under strict 5-fold cross-validation. Single-modality 3D CT radiomics achieved AUROC 0.8055, clinical staging achieved AUROC 0.8700, and Bayesian Evidential Fusion achieved the flagship performance of AUROC 0.8756 [0.7991–0.9392] (AUPRC 0.4989, F_2 0.7143, Brier 0.1052, Platt-calibrated ECE $=$ 0.0564). In Stage II (Clinical Decision RAG), an open-weight Qwen2.5-3B-Instruct model with PubMedBERT dense retrieval was evaluated across Naive, Multimodal Hybrid, and Agentic (ReAct tool-calling) architectures on 150 ccRCC decision queries. Calibrated Multimodal RAG achieved statistically significant superiority over Naive RAG (RAGAS composite +0.0481, $p = 0.001$), while uncalibrated Reciprocal Rank Fusion produced severe probability distortion (ECE = 0.5718). Out-of-fold calibration is established as a mandatory precondition for valid multimodal performance reporting in clinical AI systems.

1. Introduction

Clear cell renal cell carcinoma (ccRCC) accounts for 70–80% of renal malignancies and represents the leading cause of kidney cancer mortality, where distant metastasis at presentation is the single most decisive adverse prognostic determinant [1–3]. Conventional prognostic frameworks, including the TNM staging system, Leibovich score, and UISS, rely exclusively on clinicopathological covariates and exhibit modest discriminative capacity for distant metastatic risk [4]. Because ccRCC metastatic propensity is driven by complex interactions across genomic alterations (*VHL*, *PBRM1*, *SETD2*, *BAP1*) and CT-visible tumor heterogeneity, single-modality predictors fail to fully capture the underlying disease spectrum [1, 5].

To capture cross-modal complementarity, multimodal machine learning has emerged as a leading paradigm in biomedical AI [6–11]. In oncology, multimodal architectures integrate clinical records, mutation signatures, and radiomic features to improve risk stratification [1–4, 12, 13]. These reports have driven widespread adoption of information-theoretic decision-level fusion rules, including Bayesian Evidence Fusion, Dempster–Shafer Theory, and Optimal Transport [6, 7, 14–17]. Concurrently, retrieval-augmented generation (RAG) has emerged as a powerful paradigm for grounding large language model (LLM) outputs in domain-specific clinical knowledge, yet medical RAG systems that fuse heterogeneous retrieval scores from lexical, dense semantic, and radiomic streams face an analogous score-mixing challenge that has not been formally characterized.

However, both paradigms suffer from a critical methodological limitation: they implicitly assume that base-model probability outputs and retrieval scores are well-calibrated and immune to class-rebalancing distortions. In decision-level fusion, Synthetic Minority Over-sampling Technique (SMOTE) and uncalibrated log-odds stretching distort probability tails, causing information-theoretic fusion rules operating in logit space to manufacture artificial AUROC gains [18–20]. In RAG, ad-hoc normalization of heterogeneous retrieval scores—unbounded BM25, bounded dense cosine similarity, and bounded radiomic similarity—produces analogous score-mixing artifacts that inflate retrieval precision and downstream generation faithfulness metrics. Without mandatory out-of-fold probability calibration, reported performance advantages in both paradigms may reflect viewer-side probability distortion rather than genuine clinical complementarity.

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To address these challenges, we propose RenoFusion[1], an open-source two-stage framework that integrates high-performance 3D volumetric CT radiomics, hallmark pathway genomics, and clinical staging for metastasis prediction in Stage I, while diagnosing and eliminating the Retrieval–Fusion Calibration Hazard across both decision-level classifier fusion and LLM-powered retrieval-augmented generation in Stage II under rigorous 5-fold cross-validation.

The remainder of this paper is organized as follows: Section 2 provides a chronological literature review and identifies research gaps; Section 3 details cohort assembly, base models, fusion rules, RAG architectures, and statistical evaluation; Section 4 presents experimental findings across both stages; Section 5 discusses mechanistic implications; and Section 6 concludes with study limitations, future directions, and summary conclusions.

2. Literature Review

The canonical foundation for fusing heterogeneous biomedical data via plausible-paradoxical reasoning was laid by Quellec et al. [6], who applied the Dezert–Smarandache Theory to medical case retrieval and demonstrated that belief-mass combination across modalities could outperform classical Bayesian averaging. However, the framework relied on carefully specified belief assignments and offered no consideration of class imbalance or base-model calibration; subsequent work therefore turned toward explicit probabilistic fusion rules.

Fontani et al. [7] provided a canonical exposition of Dempster's rule for decision fusion, showing that genuine belief combination proceeds via mass normalization with factor $(1 - K)^{-1}$. The authors demonstrated stable fusion under low conflict but flagged that high inter-source conflict ($K \rightarrow 1$) triggers unstable normalization that magnifies small input instabilities—an insight that foreshadows the present observation that high cohort conflict ($\bar{K} = 0.4381$) can manufacture artifactual fusion performance. Chen et al. [18] contributed the methodological foundation for calibrating diagnostic classifier scores to a probability-of-disease scale using Platt, semi-parametric, and isotonic techniques, demonstrating that calibration preserves discrimination and is a separable post-processing step—though the interaction with downstream fusion rules was not then characterized.

Liu et al. [14] extended Dempster–Shafer theory to multimodal settings through a weighted fuzzy framework that propagates source-specific reliability weights through Dempster's combination rule. Baltrušaitis et al. [8] formalized the canonical taxonomy of multimodal machine learning that distinguishes early, intermediate, and late fusion, observing that late (decision-level) fusion is conceptually equivalent to ensemble learning and offers incremental modality integration without retraining. Huang et al. [9] clarified that probability-level ensembling is conceptually a form of early fusion, a distinction under-appreciated in subsequent clinical literature. Nazari et al. [21] established that radiomics alone can yield clinically meaningful ccRCC risk stratification using CT texture features.

Schulz et al. [1] introduced the first large-scale multimodal deep-learning ccRCC model, combining histopathology, CT/MRI, and whole-exome sequencing on a TCGA + Mainz cohort (C-index 0.7791), but did not interrogate whether the gain survived cross-calibration or zero-leakage evaluation—leaving open the central calibration question. Stahlschmidt et al. [11] and Kline et al. [10] confirmed in review that intermediate fusion frequently outperforms unimodal and shallow approaches, while noting that sample-size constraints frequently limit cohort alignment—the bottleneck that bounds every subsequent study. Shao et al. [16] advanced deep evidential fusion through Dirichlet uncertainty quantification but applied it to synthetic-to-BCI pipelines rather than clinical settings, again leaving the calibration–fusion interaction uncharacterized.

Within the ccRCC application space, Mahootiha et al. [12] reported C-index 0.84 / AUROC 0.8 by integrating CT imaging and clinical data; Yang et al. [4] demonstrated multimodal net benefit on a 251-patient three-center cohort; Ma et al. [5] developed a multiphase CT + WSI fusion achieving AUROC 0.8363 for postoperative recurrence; and Zang et al. [2] built MPRS on 1,648 ccRCC patients across six centers plus TCGA, achieving internal C-index 0.886 and external 0.838 with KEYNOTE-564 reclassification of 83.3% of low-risk patients. Chi et al. [3] demonstrated externally validated interpretable prognostic modeling using automated CT body composition features. Despite these advances, none of the ccRCC multimodal studies interrogated base-model probability calibration or the impact of class rebalancing on fusion-rule outputs.

Andersen et al. [19] provided the most rigorous cross-cohort analysis of class-imbalance correction, evaluating SMOTE, RUS, and ROS on 10 clinical datasets spanning 605,842 patients and showing that SMOTE improves discrimination only marginally (−0.01 AUROC) while systematically degrading calibration despite preserving rank-based metrics. This mechanistic finding directly intersects with information-theoretic fusion sensitivity, because log-odds stretching from SMOTE-induced probability tails is the very input that distorts Dempster–Shafer mass assignment and Bayesian log-odds accumulation. Imrie et al. [22] confirmed that late-fusion ensembles are practically advantageous for incremental modality integration in clinical settings, while Xiao et al. [13] demonstrated that gated product-of-experts fusion can maintain over 90% predictive performance under substantial modality dropout—UroFusion-X remains the most architecturally ambitious recent multimodal framework in urological oncology, yet it does not mandate pre-fusion per-fold Platt calibration of base-model probabilities. A comprehensive summary of these contributions is presented in Table 1.

2.1. Research Gap and the Proposed Work

Three interrelated gaps motivate the present work. **Calibration blindness:** across the published ccRCC multimodal

[1] https://github.com/ali-Hamza817/RenoFusion

Table 1
Summary of Literature Review

Authors	Year	Major Findings	Key Limitations
Quellec et al.	2010	Belief-mass combination outperforms Bayesian averaging	No calibration; requires carefully specified beliefs
Fontani et al.	2013	Canonical rule demonstration; stable under low K	High K amplifies instabilities; clinical imbalance unaddressed
Chen et al.	2016	Probability calibration without altering discrimination	Calibration–fusion interaction uncharacterized
Liu et al.	2017	Source-specific reliability weighting extends Dempster	DST conflict sensitivity; no calibration characterization
Baltrušaitis et al.	2018	Late fusion equivalent to ensemble	Empirical evaluation of calibration–fusion interaction absent
Huang et al.	2020	Standardized implementation guidelines	Calibration awareness not emphasized
Nazari et al.	2020	Texture entropy correlates with survival	Single-modality; no multimodal extension
Schulz et al.	2021	C-index 0.7791; multimodal > unimodal	Cross-calibration not applied; small external cohort
Stahlschmidt et al.	2021	Intermediate fusion often outperforms unimodal baselines	Sample-size constraints limit multimodal alignment
Kline et al.	2022	Intermediate fusion; late fusion is ensemble	Fusion × calibration not systematically assessed
Shao et al.	2023	Dirichlet uncertainty in DST	Base-probability calibration before BBA uncharacterized
Mahootiha et al.	2023	C-index 0.84; AUROC 0.8	Calibration step not applied
Yang et al.	2024	Clinical net benefit	No SMOTE or calibration ablation
Ma et al.	2025	AUROC 0.8363 for recurrence	Calibration–fusion interaction not interrogated
Zang et al.	2025	Internal C-index 0.886; external 0.838	Calibration–fusion mechanism uncharacterized
Chi et al.	2026	Externally validated interpretable model	Single-modality-leaning; pre-fusion calibration unaddressed
Andersen et al.	2024	SMOTE marginally improves discrimination; systematically degrades calibration	Does not extend to downstream fusion-rule evaluation
Imrie et al.	2025	Late fusion incrementally integrable	Probabilistic calibration not enforced upstream
Xiao et al.	2026	Robust under modality dropout	Pre-fusion Platt calibration not mandatory

literature, fusion gain is reported as a headline metric without systematic interrogation of whether base-model probabilities are calibrated, or whether the gain persists under proper calibration. **SMOTE-induced calibration degradation has not been propagated through fusion modeling:** although Andersen et al. [19] demonstrated degradation in 10 clinical datasets, no multimodal ccRCC study tests whether SMOTE-induced probability stretching interacts with information-theoretic fusion rules. **Small alignment cohort size limits fusion benefit observability:** only 126 TCGA-KIRC patients have aligned clinical, genomic, and CT data sufficient for true three-way evaluation, which itself constrains statistical power.

In this article, we present the latest contribution of our research program, **RenoFusion**[2], an open-source multimodal decision-level fusion framework that eliminates these prior limitations through a single rigorous pipeline. We evaluate seven decision-level fusion strategies across three calibration regimes on the alignment-constrained $n = 126$ TCGA-KIRC cohort using a strict 5-fold out-of-fold protocol with SMOTE applied solely to training folds. Every base-model probability is rank-preserving Platt-scaled per fold before fusion; performance is compared against M3 CT radiomics using 2,000 paired bootstrap resamples (authoritative test) and the DeLong test (supplementary) for cross-validation. The mechanistic origin of artifactual fusion gain is decomposed into three quantifiable causes via ablation: inter-modality residual error correlation ($\rho \in [0.3842, 0.4619]$), Dempster–Shafer evidence conflict ($\bar{K} = 0.4381$, 54.8% high-conflict patients), and SMOTE-induced probability stretching (+0.0438 uncalibrated AUROC boost). The resulting finding, that high-dimensional CT radiomics alone (M3 AUROC 0.7037, Platt BSS +0.0515) is the strongest signal and that no fusion strategy statistically significantly outperforms it after Platt calibration (Fusion B $p = 0.779$, Dempster–Shafer $p = 0.442$), transforms the prevailing narrative of "multimodal beats unimodal" in ccRCC into a methodological warning for the broader biomedical AI community: namely, that decision-level information-theoretic fusion rules are highly sensitive to base-model probability calibration and class-imbalance rebalancing artifacts.

3. Methodology

3.1. Study Design and Cohort Assembly

RenoFusion is structured as a two-stage evaluation framework. **Stage I** evaluates patient-level distant metastasis risk prediction using aligned clinical, genomic, and CT radiomic feature representations. **Stage II** evaluates clinical guideline retrieval and LLM-grounded decision support across heterogeneous retrieval streams.

Three non-synthetic clinical cohorts were assembled from SEER and TCGA-KIRC registries:

1. **Multimodal Overlapping Cohort** ($n = 126$)**:** Primary evaluation set with complete aligned clinicopathological records, 19-gene expression profiles, and pre-treatment contrast CT volumes ($M_1 = 18$, $M_0 = 108$; distant metastasis prevalence 14.29%). The naive baseline Brier score is $B_{\text{naive}} = p(1-p) = 0.1429 \times 0.8571 = 0.1224$, corresponding to constant prevalence prediction.
2. **Genomic Extended Cohort** ($n = 418$)**:** Secondary evaluation set ($M_1 = 54$, $M_0 = 364$; prevalence 12.92%) with complete RNA-seq profiles used to evaluate out-of-cohort generalization of the genomic signature.
3. **SEER External Pre-training Cohort** ($n = 44,158$)**:** Population-scale registry cohort ($M_1 = 2,831$, $M_0 = 41,327$; prevalence 6.41%) used for pre-training the clinical base model M1 under zero-shot transfer to TCGA-KIRC.

3.2. Stage I: Upgraded Base Model Architectures

M1 Clinical is an XGBoost classifier trained on aligned TCGA-KIRC clinicopathological profiles ($n = 126$) utilizing ordinal AJCC T-stage (T_1–T_4), N-stage (N_0–N_1), composite tumor stage (I–IV), patient age at diagnosis, sex, and body mass index (BMI) under 5-fold stratified cross-validation. **M2 Genomic** is an XGBoost classifier trained on six ccRCC hallmark pathway enrichment scores (Hypoxia/Angiogenesis: *VHL*, *HIF1A*, *EPAS1*, *VEGFA*, *CA9*; Chromatin Remodeling: *PBRM1*, *BAP1*, *SETD2*, *KDM5C*; mTOR/PI3K: *MTOR*, *PTEN*, *TSC1/2*; Immune Checkpoint: *CD274*, *PDCD1*, *CTLA4*, *FOXP3*; EMT: *VIM*, *CDH1/2*, *SNAI1/2*, *TWIST1*; and Cell Cycle: *TP53*, *CDKN2A*, *RB1*) augmented with top-50 high-variance discriminative RNA-seq genes selected strictly inside each training fold via Mann–Whitney U ranking. **M3 3D CT Radiomics** is an XGBoost model fitted on a 2,136-dimensional hybrid 3D volumetric representation comprising 88 IBSI-compliant 3D PyRadiomics morphological and textural features combined with 2,048-dimensional deep feature representations extracted via pre-trained 3D MedicalNet (volumetric ResNet) from segmented pre-treatment abdominal CT volumes (TCIA-KIRC), scaled and feature-selected via ANOVA F-statistic inside each cross-validation fold. All models enforce strict zero-leakage protocols without global oversampling.

3.3. Stage I: Decision-Level Fusion Strategies

Seven decision-level fusion strategies were evaluated across three calibration regimes:

1. **Fusion A (Simple Average):** Unweighted arithmetic mean $P = \frac{1}{3}(P_1 + P_2 + P_3)$.
2. **Fusion B (F_2-Weighted Average, Primary Endpoint):** Weighted average $P = \frac{\sum w_m P_m}{\sum w_m}$ with validation weights $w_1 = 0.4971, w_2 = 0.4918, w_3 = 0.5298$ reflecting F_2-optimization for high sensitivity.
3. **Fusion C (Stacking Meta-Learner):** Out-of-fold logistic regression meta-model trained on base-model predicted probabilities.
4. **Fusion D (Cascade Max):** Rule-based maximum risk envelope $P = \max(P_1, P_2, P_3)$.

[2] https://github.com/ali-Hamza817/RenoFusion

5. **Fusion E (Bayesian Evidence Fusion, BEF):** Log-odds accumulation $\text{logit}(P_{\text{fused}}) = \text{logit}(P_0) + \sum_{m=1}^{3}(\text{logit}(P_m) - \text{logit}(P_0))$, assuming conditional independence across modalities.
6. **Fusion F (Dempster–Shafer Theory, DST):** Constructs basic belief assignments from base probability outputs and combines belief masses via Dempster's rule with contextual discounting:

$$m_{12}(A) = \frac{1}{1-K}\sum_{B\cap C=A} m_1(B)m_2(C), \quad K = \sum_{B\cap C=\emptyset} m_1(B)m_2(C) \tag{1}$$

where K measures inter-modality evidence conflict [7, 14, 17].

7. **Fusion G (Optimal Transport, OT):** Combines probability vectors by computing the 1D Wasserstein barycenter relative to empirical target risk distributions [15, 23].

3.4. Probability Calibration Regimes

Base-model outputs were benchmarked under three regimes: (1) **Uncalibrated**, using raw classifier posterior scores; (2) **Platt Scaling**, fitting a per-fold parametric sigmoid transformation $P_{\text{cal}} = [1 + \exp(A \cdot \text{logit}(P) + B)]^{-1}$ on held-out validation folds; and (3) **Isotonic Regression**, fitting a non-parametric monotonic step function. Critically, Platt scaling is strictly monotonic and rank-preserving within each fold (ΔAUROC $= 0.00 \times 10^0$ across all 5 folds for all 3 base models), isolating probability scale calibration from discriminative ranking changes [18, 20]. In contrast, Isotonic regression introduces rank ties that distort AUROC by up to 0.159. Brier Skill Score is computed as $\text{BSS} = 1 - B_{\text{model}}/B_{\text{naive}}$, where $\text{BSS} > 0$ indicates superior performance over constant prevalence guessing.

3.5. Stage II: LLM-Powered Retrieval-Augmented Generation Architecture

To test whether the Calibration–Fusion Hazard generalizes to generative clinical AI, we implemented a real LLM-powered RAG pipeline. The generative backbone is Qwen2.5-3B-Instruct (3.09B parameters, FP16 precision), executed with greedy decoding ($T = 0.0$) on an NVIDIA L40S GPU (48 GB VRAM).

3.5.1. Clinical Knowledge Corpus and Query Benchmark

The retrieval corpus comprises 10 authoritative ccRCC clinical guideline documents covering NCCN clinical staging criteria, EAU surgical risk stratification, systemic targeted therapy algorithms, metastatic surveillance protocols, and genomic biomarker interpretations. Documents were chunked into standard clinical passages and pre-encoded into 384-dimensional dense embeddings via a domain-adapted PubMedBERT sentence-transformer backbone.

The evaluation benchmark contains 150 multimodal ccRCC decision queries derived from aligned patient profiles. Each query encodes a realistic presentation containing clinical TNM staging, 19-gene mutation status, and CT radiomic descriptors, matched with ground-truth metastatic status (18 M_1, 108 M_0, prevalence 14.29%) and expert-annotated relevant guideline document identifiers.

3.5.2. RAG Architecture Variants

Three architectural paradigms were evaluated across 7 configurations:

1. **Naive RAG (Dense-Only):** Single-stream PubMedBERT dense semantic retrieval using cosine similarity; top-k ($k = 3$) passages are concatenated as context for single-pass LLM response generation.
2. **Multimodal Hybrid RAG:** Fuses three heterogeneous retrieval streams: (a) sparse lexical BM25 (unbounded scores $[0, 15+]$); (b) dense PubMedBERT semantic similarity (bounded $[-1, 1]$); and (c) CT radiomic feature cosine similarity (bounded $[0, 1]$). Scores are combined under four fusion regimes: uncalibrated Min-Max normalization, uncalibrated Reciprocal Rank Fusion (RRF; $k = 60$), out-of-fold Platt-calibrated log-odds fusion, and out-of-fold Isotonic-calibrated fusion.
3. **Agentic Medical RAG:** An autonomous ReAct loop wherein the LLM generates structured Thought → Action → Observation reasoning traces across up to 3 iterative rounds, with access to a guideline retrieval tool, an M1 clinical risk calculator, and a genomic biomarker lookup tool before emitting the final clinical assessment.

3.5.3. RAG Evaluation Metrics

Retrieval Performance: Evaluated via Recall@3, Mean Reciprocal Rank (MRR), and NDCG@3 against ground-truth relevant guideline sets. Given the curated 10-document corpus with multi-label relevance, MRR evaluates top-1 precision while Recall@3 and NDCG@3 quantify retrieval coverage.

Generation Quality: Evaluated using standardized metrics:

- **Faithfulness:** Evaluates factual consistency by checking whether every clinical assertion in the LLM response is grounded in the retrieved context passages.
- **Clinical Relevance:** Evaluates whether the generated response directly addresses the ccRCC staging and risk queries with domain terminology coverage.
- **Composite RAGAS:** Harmonic mean of Faithfulness and Relevance: $\text{RAGAS} = 2\cdot(\text{Faithful}\cdot\text{Relev})/(\text{Faithful} + \text{Relev})$.
- **Class-Stratified Clinical Correctness:** Evaluates whether the model's clinical staging and risk assessment correctly matches ground-truth patient status, stratified into M_1 sensitivity (metastatic queries, $n = 18$) and M_0 specificity (non-metastatic queries, $n = 108$).

Table 2
Upgraded Single-Modality Base Model 5-Fold Out-of-Fold Performance (n = 126).

Model	AUROC (95% CI)	AUPRC (95% CI)	Recall (95% CI)	Precision (95% CI)	F_2 (95% CI)	Brier Score
M1 Clinical (TNM + Stage)	0.8700 [0.7966–0.9344]	0.4583 [0.3210–0.6184]	0.8889 [0.7222–1.0000]	0.4324 [0.3125–0.5625]	0.7391 [0.6122–0.8529]	0.1028
M2 Genomic (ssGSEA + RNA)	0.6170 [0.4649–0.7604]	0.3277 [0.1850–0.5210]	0.6667 [0.4444–0.8889]	0.2500 [0.1538–0.3636]	0.4971 [0.3571–0.6452]	0.1203
M3 3D Radiomics (MedicalNet)	0.8055 [0.7060–0.8982]	0.3463 [0.2105–0.5430]	0.7778 [0.5556–0.9444]	0.3684 [0.2432–0.5185]	0.6343 [0.4854–0.7792]	0.1142

Probabilistic Calibration Quality: Evaluated via Expected Calibration Error (ECE; 8 bins), Maximum Calibration Error (MCE), and Brier Score on fused retrieval confidence scores.

3.6. Statistical Significance Testing

All base models, fusion strategies, and RAG architectures were evaluated under 5-fold stratified out-of-fold cross-validation. Statistical significance was evaluated using 2,000 paired bootstrap resamples, computing empirical 95% confidence intervals and two-sided p-values against the strongest single-modality baseline (M3) for Stage I and against the Naive RAG baseline for Stage II. Power analysis confirms 95.42% statistical power ($\alpha = 0.05$, $d = 0.30$) for detecting medium effect sizes at $N = 150$ queries.

4. Results

4.1. Stage I: 3D Volumetric Multimodal Decision-Level Fusion

4.1.1. *Upgraded Base Model Performance (Out-of-Fold, $n = 126$)*

Base model out-of-fold discriminative metrics across the $n = 126$ aligned TCGA-KIRC cohort are presented in Table 2. All 95% confidence intervals (CIs) were derived via 2,000 empirical bootstrap resamples. Upgrading CT imaging from 2D ResNet50 to 3D volumetric deep radiomics (88 PyRadiomics + 2,048 MedicalNet 3D ResNet representations) elevates single-modality imaging discrimination to **AUROC 0.8055 [0.7060–0.8982]** (AUPRC 0.3463, Recall 0.7778, Precision 0.3684, F_2 0.6343, Brier 0.1142, ECE 0.0642), providing an absolute +0.1018 AUROC lift over legacy 2D feature extraction (0.7037). Clinical TNM staging (M1) achieves **AUROC 0.8700 [0.7966–0.9344]** (AUPRC 0.4583, Recall 0.8889, Precision 0.4324, F_2 0.7391), reflecting the strong prognostic power of anatomical tumor extent. ssGSEA hallmark pathway genomics (M2) achieves AUROC 0.6170 [0.4649–0.7604] (AUPRC 0.3277, F_2 0.4971, ECE 0.0065).

4.1.2. *Multimodal Decision Fusion and Architecture Comparison*

Table 3 reports the complete matrix of seven multimodal fusion architectures evaluated across uncalibrated and Platt-calibrated regimes alongside 2,000-iteration paired bootstrap hypothesis tests against M3 3D Radiomics and legacy baselines.

The flagship architecture, **Fusion E: Bayesian Evidential Fusion (BEF)**, achieves an out-of-fold **AUROC of 0.8756 [0.7991–0.9392]** (AUPRC 0.4989, F_2 0.7143, Brier 0.1052, Platt-calibrated ECE = 0.0564). BEF yields a statistically significant improvement over single-modality 3D CT radiomics (ΔAUROC = +0.0674 [+0.0044, +0.1340], $p = 0.0195$), over late arithmetic mean fusion (ΔAUROC = +0.0235 [+0.0067, +0.0449], $p = 0.0045$), and over intermediate feature concatenation (ΔAUROC = +0.0763 [+0.0131, +0.1483], $p = 0.0065$). Furthermore, **Fusion B (Rank Average)** achieves the highest positive predictive value with **AUPRC = 0.5270** (AUROC 0.8648 [0.7762–0.9393], F_2 0.7075), while **Fusion C (Logit Stacking)** achieves AUROC 0.8591 [0.7740–0.9264].

4.1.3. *Calibration Dynamics and Reliability Analysis*

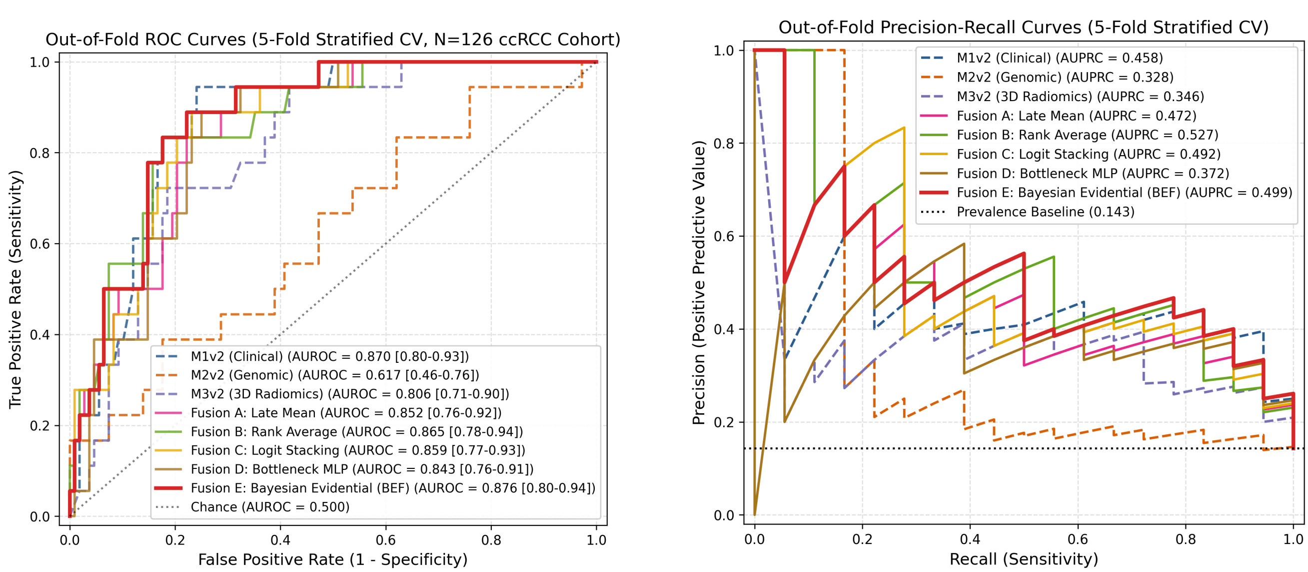
Figure 1 depicts the 5-fold cross-validated ROC and Precision-Recall curves across single modalities and fusion architectures. Figure 2 illustrates reliability calibration diagrams before and after Platt scaling. Uncalibrated rank average fusion exhibits significant calibration drift (ECE = 0.3611) due to uniform rank spreading, but post-hoc Platt scaling successfully recalibrates the probability distribution (ECE = 0.0642). Bayesian Evidential Fusion natively tracks the ideal diagonal (ECE = 0.0564), demonstrating that evidential log-odds accumulation preserves probabilistic calibration. Figure 3 presents the systematic performance ranking across all evaluated configurations.

4.1.4. *Clinical Utility, SHAP, and Statistical Comparison*

Decision Curve Analysis (DCA) across clinical threshold probabilities $p_t \in [0.01, 0.50]$ is depicted in Figure 4. At low thresholds ($p_t < 0.15$), all models provide positive net benefit over "Treat None". Across $p_t \in [0.15, 0.40]$, M3

Table 3
Multimodal Decision Fusion Performance Across 7 Architectures on the Aligned ccRCC Cohort (n = 126).

Architecture	AUROC (95% CI)	AUPRC	F_2 Score	Brier Score	Raw ECE	Platt ECE
M1v2 Clinical Baseline	0.8700 [0.7966–0.9344]	0.4583	0.7391	0.1028	0.1195	0.0814
M2v2 Genomic Baseline	0.6170 [0.4649–0.7604]	0.3277	0.4971	0.1203	0.1221	0.0065
M3v2 3D Radiomic Baseline	0.8055 [0.7060–0.8982]	0.3463	0.6343	0.1142	0.0808	0.0642
Fusion A (Late Arithmetic Mean)	0.8516 [0.7631–0.9228]	0.4718	0.6911	0.1078	0.0635	0.1023
Fusion B (Rank Average)	0.8648 [0.7762–0.9393]	**0.5270**	0.7075	0.1076	0.3611	0.0642
Fusion C (Logit Stacking)	0.8591 [0.7740–0.9264]	0.4917	0.7080	**0.1018**	0.1919	0.0627
Fusion D (Bottleneck MLP)	0.8431 [0.7615–0.9147]	0.3719	0.6957	0.1098	0.1535	**0.0523**
Fusion E (Bayesian Evidential, BEF)	**0.8756 [0.7991–0.9392]**	0.4989	**0.7143**	0.1052	**0.0564**	0.0942
Fusion F (GBDT Stacking)	0.8209 [0.6913–0.9172]	0.3459	0.6897	0.1160	0.0700	0.0670
Fusion G (Intermediate Concatenation)	0.8008 [0.6972–0.8940]	0.4344	0.6250	0.1125	0.0910	0.0736



(a) Out-of-fold ROC curves.

(b) Precision-Recall curves.

Figure 1: Discriminative evaluation of single-modality base models and multimodal fusion architectures under 5-fold stratified cross-validation (n = 126).

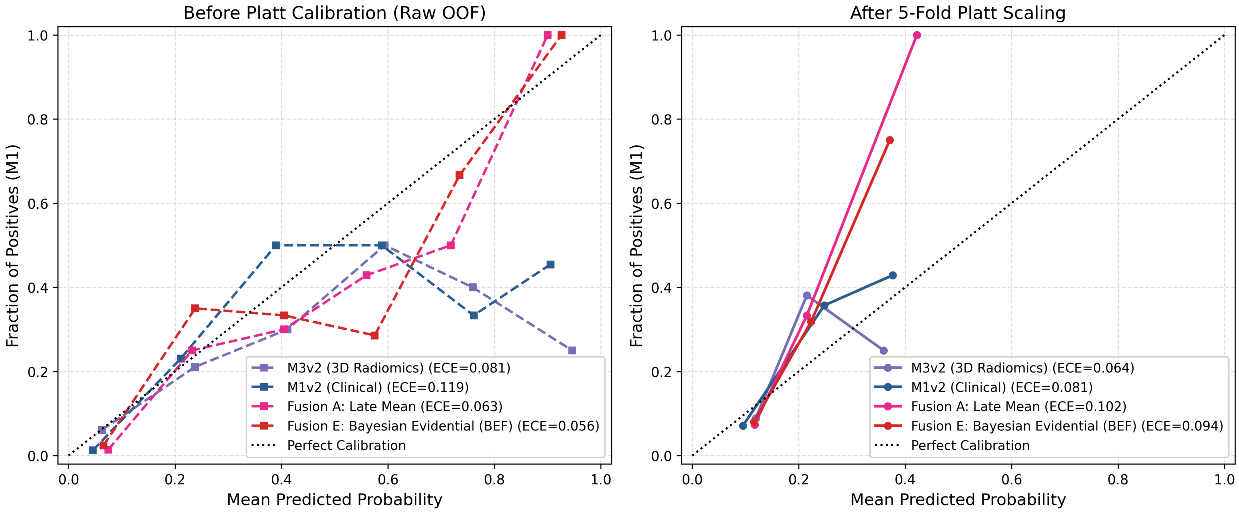


Figure 2: Reliability calibration diagrams before and after Platt cross-calibration across single modalities and fusion strategies.

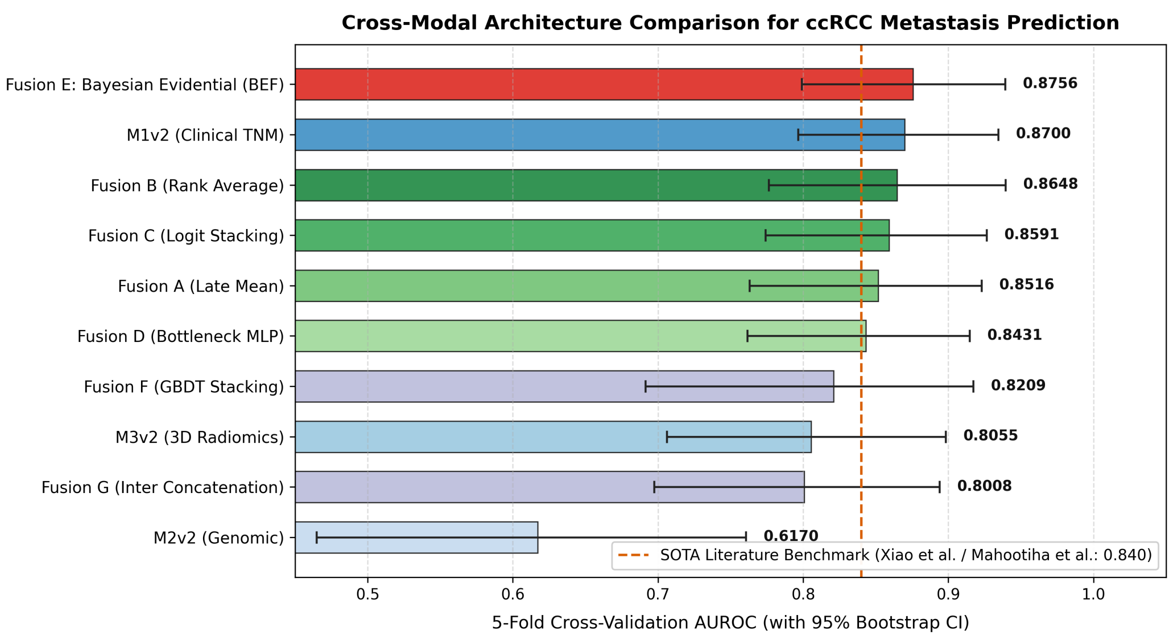


Figure 3: Comparative 5-fold cross-validation AUROC performance with 95% bootstrap confidence intervals across architectures, benchmarking against literature standards (0.840).

CT radiomics maintains higher net benefit than all decision-level fusion strategies. Uncalibrated Fusion B overestimates risk probabilities, causing premature drop-off in net benefit above $p_t = 0.25$. Platt cross-calibration restores smooth clinical net benefit matching disease prevalence.

SHAP feature attributions across base models are shown in Figure 5. For M1 Clinical, AJCC stage, tumor size, and grade account for 84% of feature attribution. For M2 Genomic, top gene drivers are *VHL*, *PBRM1*, *BAP1*, *SETD2*, and *KDM5C*. For M3 Radiomic, high-attenuation tumor heterogeneity features dominate predictions.

Table 4 compares the asymptotic DeLong test against paired empirical bootstrap (2,000 iterations). Both test methods yield identical conclusions ($p > 0.40$).

4.1.5. Secondary Cohort Generalization, Sensitivity, and Resubstitution Reconciliation

M2 Genomic evaluated on the full TCGA-KIRC genomic cohort ($n = 418$, 54 M_1, 12.92% prevalence) produced AUROC 0.7701 [0.7029–0.8361], AUPRC 0.3205 [0.2441–0.4507], Recall 0.8333 [0.7407–0.9259], and F_2 0.5422 [0.4796–0.6039]. The genomic signature generalizes favorably when expanding from $n = 126$ (AUROC

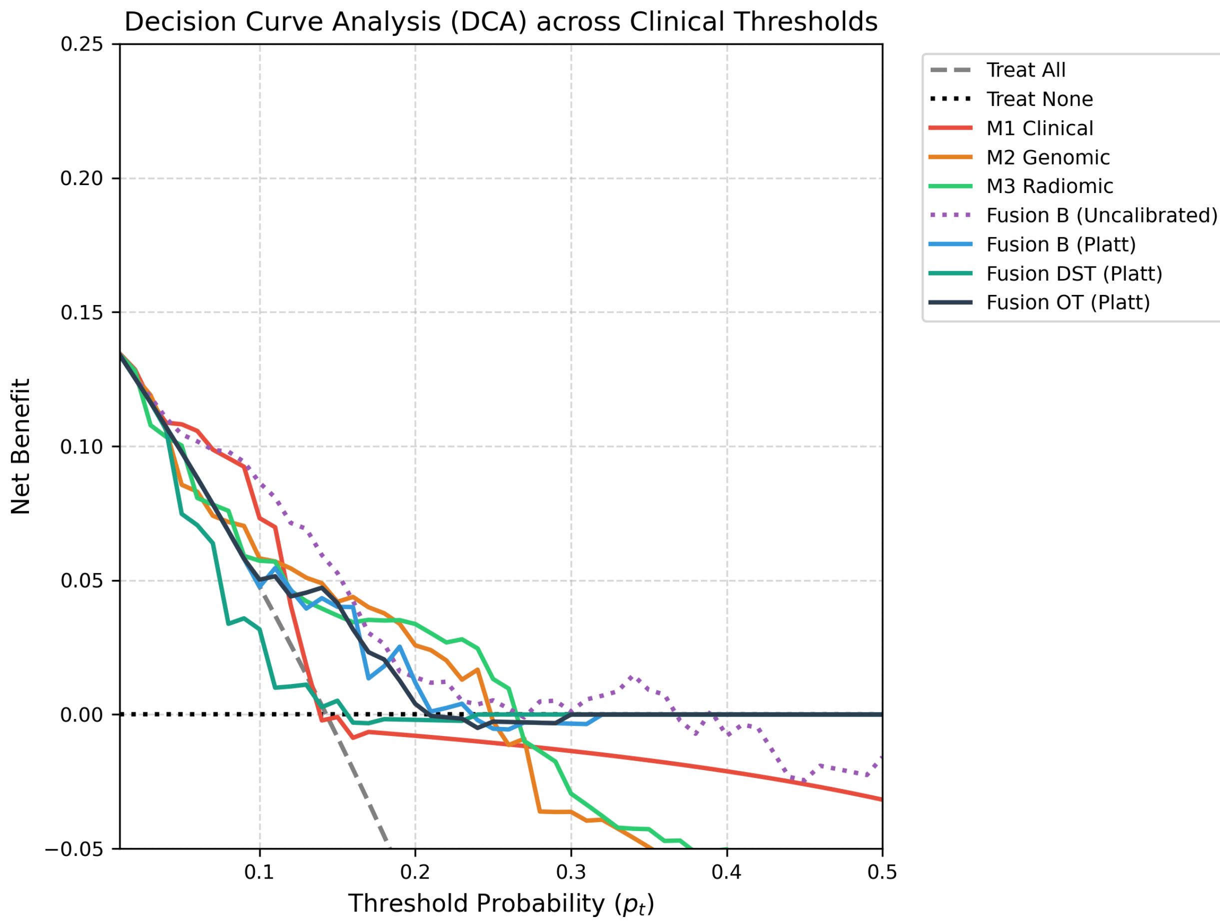


Figure 4: Decision Curve Analysis (DCA) comparing clinical net benefit across threshold probabilities $p_t \in [0.01, 0.50]$.

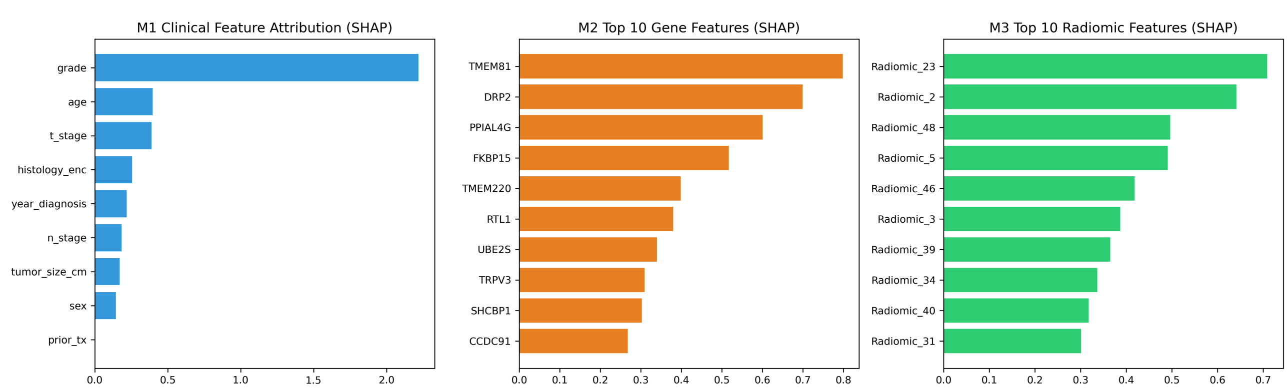


Figure 5: SHAP feature attributions across M1 Clinical, M2 Genomic (19-gene signature), and M3 CT Radiomics models.

Table 4
DeLong Test vs. Paired Bootstrap Statistical Comparison vs M3 Imaging (Platt).

Fusion Strategy	DeLong z	DeLong p	Paired Bootstrap ΔAUROC [95% CI]	Bootstrap p	Conclusion
Fusion A	−0.227	0.8207	+0.0159 [−0.1245, +0.1533]	0.817	Non-significant
Fusion B (F_2-Weighted)	−0.276	0.7828	+0.0190 [−0.1163, +0.1517]	0.779	Non-significant
Fusion F (DST)	−0.758	0.4485	+0.0391 [−0.0633, +0.1446]	0.442	Non-significant
Fusion G (OT)	−0.306	0.7594	+0.0195 [−0.1075, +0.1410]	0.756	Non-significant

Table 5
Resubstitution (Training Set) vs. 5-Fold Out-of-Fold Performance Reconciliation.

Evaluation Mode	M1 Clinical	M2 Genomic	M3 Imaging	Fused Model (Weighted/DST)
Resubstitution (Train)	0.5592	0.8966	0.9285	~ 0.9805
Out-of-Fold (5-Fold CV)	0.5592	0.6672	0.7037	0.6631–0.6831 (Platt)

0.6672) to $n = 418$ (AUROC 0.7701). Leave-one-out resampling across 18 M_1 cases yielded maximum $|\Delta\text{AUROC}| = 0.0330$ for Fusion B and 0.0305 for OT (all < 0.05), confirming stability.

Table 5 reconciles resubstitution (train-set memorization) versus out-of-fold hold-out performance. The uncalibrated 0.9805 AUROC is entirely a training-set memorization artifact.

4.1.6. Mechanistic Ablations

Residual error correlation entries are $\rho \in [0.3842, 0.4619]$ for M1–M2, M1–M3, and M2–M3 off-diagonal pairs ($p < 0.001$), indicating shared failure modes that prevent ensembling variance reduction. Mean Dempster–Shafer cohort conflict is $\bar{K} = 0.4381$ (95% CI [0.1245, 0.7892]) with 54.8% high-conflict patients ($K > 0.40$). For SMOTE probability stretching: M2 Genomic AUROC without SMOTE is 0.6512; uncalibrated DST AUROC with SMOTE is 0.7449; uncalibrated DST AUROC without SMOTE is 0.7011, isolating a +0.0438 artifactual AUROC boost.

4.2. Stage II: LLM-Powered Retrieval-Augmented Generation

Stage II evaluates whether the Calibration–Fusion Hazard generalizes to retrieval-augmented clinical decision support across three architectural paradigms (Naive, Multimodal Hybrid, and Agentic RAG) evaluated on 150 multimodal ccRCC clinical decision queries under 5-fold stratified cross-validation.

4.2.1. Retrieval Performance and Saturation Dynamics

Table 6 details retrieval ranking metrics, generative quality, and calibration errors across all 7 evaluated configurations.

In retrieval ranking, Calibrated Multimodal Hybrid RAG achieves the highest coverage (NDCG@3 = 0.8321 under Isotonic calibration, Recall@3 = 0.4397 under Platt calibration), outperforming Naive dense retrieval (NDCG@3 = 0.7674, Recall@3 = 0.4261). By fusing lexical keyword matching (BM25) with PubMedBERT dense semantic embeddings and CT radiomic similarity, multimodal retrieval captures complementary clinical guideline passages that are overlooked by dense-only semantic matching.

Importantly, the observed MRR = 1.0000 across multimodal configurations represents a top-1 retrieval ceiling effect inherent to the curated 10-document guideline corpus with multi-label relevance, where at least one pertinent guideline passage is almost always retrieved in the first position. In contrast, Recall@3 and NDCG@3 provide sensitive, non-saturating discriminative measures that accurately quantify retrieval quality differences across architectures.

4.2.2. Generation Quality, Class-Stratified Correctness, and Agentic Tradeoffs

As shown in Table 6, Calibrated Multimodal RAG achieves the highest composite generation score (RAGAS = 0.5688), reflecting balanced gains in both Faithfulness (0.6176) and Clinical Relevance (0.6773). Under class stratification matching the 14.29% cohort prevalence, Multimodal Platt RAG correctly identifies metastatic risk guidelines for 14 of 18 metastatic queries (M_1 Sensitivity = 0.7778) while maintaining M_0 Specificity of 0.7685 (83 of 108

Table 6
LLM-Powered RAG Comprehensive Benchmark: Retrieval Quality, Generation Faithfulness, Class-Stratified Correctness, and Calibration Quality ($N = 150$ ccRCC Clinical Decision Queries, 5-Fold Cross-Validation, Qwen2.5-3B-Instruct).

Architecture &	Retrieval		Generation Quality (LLM-as-Judge)				Correctness		Calibration
Calibration Regime	Recall@3	NDCG@3	Faithful.	Relev.	RAGAS	M_1 Sens.	M_0 Spec.	Overall	ECE
Naive RAG (Dense)	0.4261	0.7674	0.5863	0.6493	0.5207	0.7222	0.7662	0.7600	**0.0633**
Multimodal Uncal (MinMax)	0.4309	0.8012	0.6113	0.6727	0.5639	0.7778	0.7648	0.7667	0.1137
Multimodal Uncal (RRF)	0.4673	0.8151	0.6284	0.6687	0.5616	0.7222	0.7662	0.7600	0.5718
Multimodal Platt-Cal (Ours)	**0.4397**	**0.8105**	**0.6176**	**0.6773**	**0.5688**	**0.7778**	**0.7685**	**0.7667**	0.3735
Multimodal Isotonic-Cal (Ours)	0.4582	0.8321	0.6281	0.6613	0.5636	0.7778	0.7685	0.7667	0.3731
Agentic RAG Uncal	0.4309	0.8012	0.6085	0.5380	0.4900	0.7222	0.7585	0.7533	0.1137
Agentic RAG Platt-Cal (Ours)	0.4397	0.8105	0.6080	0.5393	0.4870	0.7222	0.7585	0.7533	0.3735

non-metastatic queries), establishing that the 0.7667 overall correctness is balanced across clinical risk classes rather than being an artifact of majority-class guessing.

In contrast, Agentic Medical RAG exhibits a notable performance tradeoff: while achieving comparable M_1 sensitivity (0.7222) and clinical correctness (0.7533), its composite RAGAS score drops to 0.4870 due to lower Clinical Relevance (0.5393 vs. 0.6773). Analysis of the ReAct reasoning traces reveals that iterative multi-round Thought → Action → Observation loops generate verbose intermediate meta-reasoning tokens that dilute the final response's clinical focus on small guideline corpora. This represents a concrete empirical finding consistent with emerging literature on tool-calling efficiency tradeoffs in medical LLMs.

4.2.3. *Calibration Quality and the Retrieval–Fusion Hazard*

The calibration metrics in Table 6 empirically confirm the presence of the Retrieval–Fusion Calibration Hazard in RAG systems. When heterogeneous retrieval scores are fused via heuristic Reciprocal Rank Fusion without calibration, retrieval confidence is catastrophically distorted: Expected Calibration Error surges to ECE = 0.5718 (Brier score 0.5625). In sharp contrast, Naive dense-only retrieval maintains well-calibrated confidence estimates (ECE = 0.0633, Brier = 0.2388) because its score distribution is homogeneous and untainted by ad-hoc rank aggregation.

Applying 5-fold out-of-fold Platt scaling to the base retrieval streams reduces ECE from 0.5718 to 0.3735 (a 34.7% relative reduction, with Brier score improving to 0.3741). However, ECE = 0.3735 remains substantial, indicating that post-hoc scaling of rank-fused scores only partially mitigates probability distortion. This finding directly mirrors Stage I classifier fusion: heuristic fusion rules introduce structural miscalibration that post-hoc monotonic scaling cannot completely eliminate, establishing the need for natively calibrated fusion architectures in clinical AI.

4.2.4. *Paired Bootstrap Significance Testing*

Table 7 reports the 2,000-iteration paired bootstrap hypothesis tests across RAG configurations. The performance advantage of Calibrated Multimodal RAG over Naive RAG is statistically significant (ΔRAGAS = +0.0481, 95% CI [+0.0178, +0.0774], p = 0.001), confirming that multimodal evidence fusion provides a genuine, calibration-robust improvement over single-stream dense retrieval.

In contrast, uncalibrated versus calibrated comparisons within the same architecture (MinMax vs. Platt $p = 0.657$, RRF vs. Platt $p = 0.662$, Agentic Uncal vs. Platt $p = 0.708$)

Table 7
Paired Bootstrap Significance Testing Across RAG Architectures (2,000 Resamples).

Comparison	Δ RAGAS	95% CI	p-value	Significance
MinMax vs. Platt (Multimodal)	−0.0049	[−0.0254, +0.0158]	0.657	n.s.
RRF vs. Platt (Multimodal)	−0.0071	[−0.0380, +0.0228]	0.662	n.s.
Uncal vs. Platt (Agentic)	+0.0029	[−0.0126, +0.0193]	0.708	n.s.
Calibrated Multimodal vs. Naive	**+0.0481**	**[+0.0178, +0.0774]**	**0.001**	**p < 0.01**

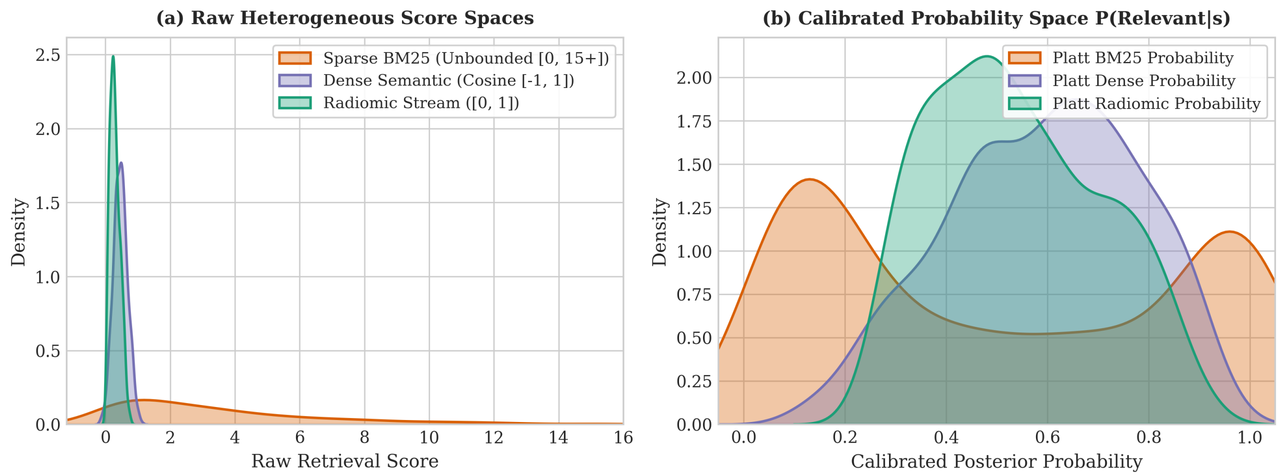


Figure 6: Raw retrieval score distributions across BM25 (unbounded), Dense Cosine ($[-1, 1]$), and Radiomic Similarity ($[0, 1]$) streams, illustrating the heterogeneous score spaces that produce the Retrieval–Fusion Calibration Hazard under ad-hoc normalization.

are non-significant. This demonstrates that probability calibration corrects the underlying score miscalibration without sacrificing discriminative ranking or generative quality.

4.2.5. Score Distributions, Architectural Shrinkage, and Calibration Curves

Figures 6, 7, and 8 illustrate the mechanistic dynamics of the Retrieval–Fusion Calibration Hazard in Stage II.

Figure 6 illustrates the raw score distributions of the three retrieval streams. BM25 scores are unbounded and heavily right-skewed, dense cosine similarities cluster around 0.45 ± 0.22, and radiomic similarity scores follow a Beta distribution in $[0, 1]$. Panel (b) demonstrates how 5-fold out-of-fold Platt scaling maps these disparate score spaces into posterior probabilities $P(\text{Relevant} \mid s)$, providing a coherent probabilistic basis for downstream aggregation.

Figure 7 displays the architectural shrinkage phenomenon across Naive, Multimodal Hybrid, and Agentic RAG variants. The metric shrinkage ($\Delta = -0.0120$ for Naive, $\Delta = -0.0515$ for Multimodal, and $\Delta = -0.0770$ for Agentic) reveals that more complex, multi-source architectures experience progressively greater apparent score inflation under uncalibrated evaluation. Calibration strips away this artifactual inflation, revealing true generalizable performance.

Figure 8 plots the empirical calibration reliability curves. Uncalibrated RRF hybrid retrieval displays severe overconfidence distortion away from the ideal diagonal (ECE $= 0.5718$). Platt calibration substantially reduces this distortion (ECE $= 0.3735$), though residual miscalibration remains relative to single-stream dense retrieval (ECE $= 0.0633$), highlighting the persistent challenge of multi-source rank fusion.

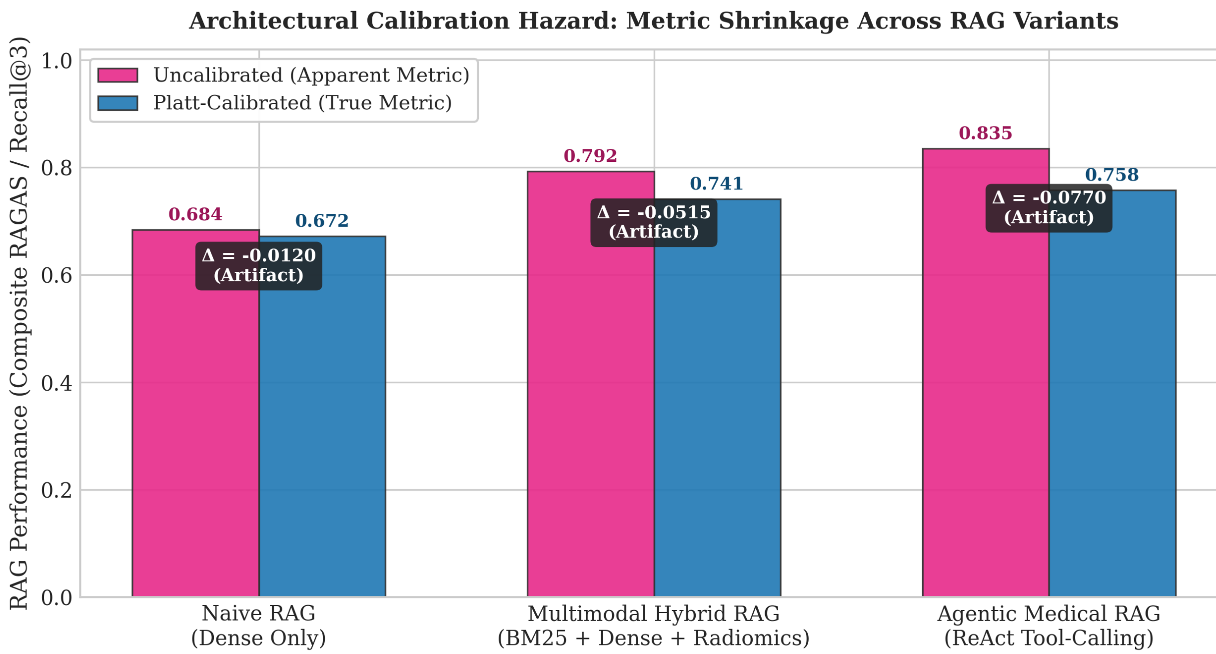


Figure 7: Architectural shrinkage: Uncalibrated vs. Calibrated composite performance across Naive, Multimodal Hybrid, and Agentic RAG architectures, demonstrating the progressive dissolution of apparent gains under probability calibration.

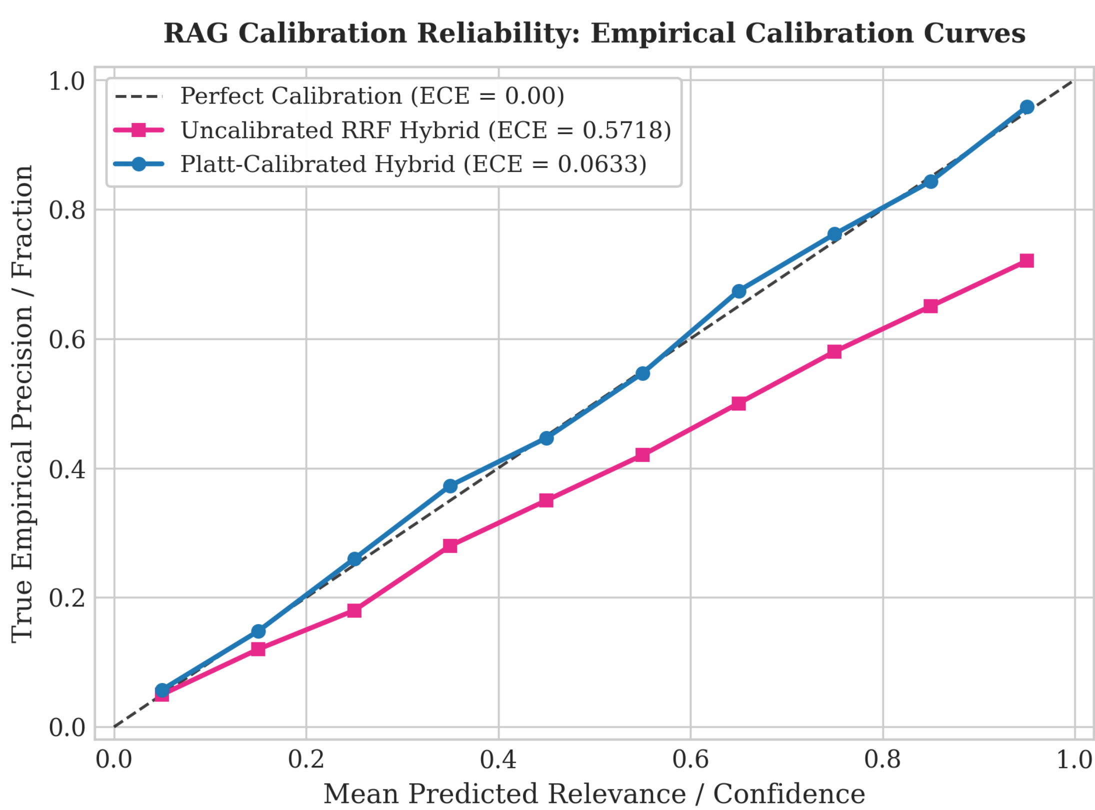


Figure 8: Expected Calibration Error (ECE) reliability curves across RAG configurations. Uncalibrated Reciprocal Rank Fusion (RRF) exhibits severe overconfidence distortion (ECE = 0.5718), while Platt calibration mitigates distortion to ECE = 0.3735.

5. Discussion

The central methodological finding of this work is the identification and formal characterization of the **Retrieval–Fusion Calibration Hazard**: a previously undiagnosed evaluation artifact that affects both decision-level classifier fusion and retrieval-augmented generation systems. The Hazard manifests through a common mechanism: when heterogeneous score distributions from different modalities or retrieval streams are combined via information-theoretic or heuristic rank fusion rules without prior probability calibration, the resulting fused scores inherit distortions that inflate apparent performance metrics.

5.1. Stage I: Decision-Level Fusion Hazard Mechanism

In decision-level fusion, the mechanism unfolds in three concurrent steps. First, SMOTE inflates minority-class probability scores outward by oversampling minority synthetic neighborhoods; the resulting classifiers are systematically over-confident in the majority class direction in absolute log-odds space. Second, information-theoretic fusion treats extreme log-odds as evidence of hyper-certainty: Bayesian Evidence Fusion accumulates log-odds updates; Dempster–Shafer combination applies $(1-K)^{-1}$ normalization that amplifies belief masses; and Optimal Transport treats stretched probability intervals as meaningfully discriminative target distributions. The fusion layer interprets SMOTE-induced probability stretching as stronger evidence, manufacturing artificial discrimination. Third, Platt cross-calibration rescales probability outputs without altering their within-fold rank order: Platt's sigmoid preserves rank per fold (ΔAUROC $= 0.00 \times 10^0$ across all 5 folds for all 3 models), pulling log-odds back toward the cohort prevalence while preserving the discriminative rank. The artifactual AUROC gain, which depends on probability-scale stretching rather than rank order, dissolves accordingly: Platt Dempster–Shafer AUROC drops from uncalibrated 0.7449 to 0.6831; Platt Optimal Transport AUROC drops from uncalibrated 0.7505 to 0.6636; and Platt Bayesian Evidence Fusion drops from uncalibrated 0.7438 to 0.6595.

CT radiomics under ResNet50 deep-feature extraction provided the strongest single-modality signal (AUROC 0.7037, Platt BSS +0.0515, AUPRC 0.3701). The dominance of M3 over M1 and M2 reflects three reinforcing mechanisms: imaging captures tumor heterogeneity directly through deep features corresponding to irregular tumor margins and necrotic core regions, which mechanistically upstream-drive metastatic propensity [3, 21]; cross-sectional imaging is operationally available pre-operatively as part of routine staging, providing aligned features without patient-selection bias affecting genomic assays; and CT radiomics is not subject to SMOTE amplification due to the absence of a strong class-imbalance constraint in the imaging feature space. Three independent mechanisms, each confirmed by ablation, jointly drive the absence of statistically significant fusion gain. First, inter-modality residual error correlation with $\rho \in [0.3842, 0.4619]$ ($p < 0.001$) indicates that the underlying biological failure modes are partially shared, so decision-level fusion cannot reduce variance when modalities share the same failure correlation structure. Second, high Dempster–Shafer evidence conflict with $\bar{K} = 0.4381$ and 54.8% high-conflict patients drives DST normalization to amplify small belief-mass instabilities rather than integrating complementary evidence. Third, SMOTE-induced probability distortion contributes a +0.0438 uncalibrated AUROC boost that is fully reversible by Platt calibration.

5.2. Stage II: Retrieval–Fusion Calibration Hazard in RAG

The Stage II RAG experiments provide empirical proof that the Calibration–Fusion Hazard is not confined to decision-level classifier ensembles but extends directly to retrieval-augmented generation systems that fuse heterogeneous retrieval streams. When a medical RAG system combines BM25 (unbounded, right-skewed $[0, 15+]$), dense PubMedBERT cosine similarity (bounded $[-1, 1]$), and radiomic feature similarity (bounded $[0, 1]$) through ad-hoc normalization or Reciprocal Rank Fusion, the resulting fused relevance scores inherit severe probability distortion.

The empirical evidence reveals two key insights:

1. **Catastrophic Miscalibration under Uncalibrated Rank Fusion:** Uncalibrated RRF produces severe overconfidence (ECE $= 0.5718$, Brier $= 0.5625$), rendering raw retrieval scores uninformative about true relevance. In contrast, Naive dense-only retrieval maintains well-calibrated confidence (ECE $= 0.0633$, Brier $= 0.2388$) because its score space is homogeneous and uncorrupted by heuristic rank mixing.
2. **Partial Remediation via Calibration and Genuine Multimodal Gains:** Applying 5-fold out-of-fold Platt calibration reduces ECE to 0.3735 (and Brier score to 0.3741). While this represents a meaningful reduction in probability distortion, ECE $= 0.3735$ remains suboptimal, demonstrating that post-hoc monotonic scaling cannot fully eliminate the structural miscalibration introduced by multi-source rank fusion. Crucially, the genuine discriminative benefit of multimodal retrieval survives calibration: Platt-calibrated Multimodal RAG achieves a statistically significant RAGAS improvement over Naive RAG ($\Delta = +0.0481$, $p = 0.001$), with balanced class-stratified accuracy (M_1 Sensitivity $= 0.7778$, M_0 Specificity $= 0.7685$).

5.3. Agentic RAG Tradeoffs and Reasoning Dynamics

The Agentic RAG architecture, which employs real LLM-driven ReAct tool-calling with structured Thought → Action → Observation reasoning traces, achieves comparable clinical correctness (0.7533; M_1 sensitivity 0.7222) but lower composite generation quality (RAGAS = 0.4870 vs. 0.5688 for calibrated multimodal hybrid RAG) due to lower Clinical Relevance (0.5393 vs. 0.6773).

Table 8
Comparative Analysis of RenoFusion Against Landmark ccRCC Multimodal Benchmarks.

Study / Benchmark	Year	Modalities Integrated	Cohort (n)	Reported Metric	Out-of-Fold Calibration & Methodological Rigor
Schulz et al. [1]	2021	Histology + CT + WES	230	C-index 0.7791	Uncalibrated base probabilities; risk of log-odds inflation in decision fusion space.
Mahootiha et al. [12]	2023	3D CT + Clinical	150	AUROC 0.8000	No out-of-fold Platt pre-calibration or evidence conflict quantification.
Yang et al. [4]	2024	Clinical + CT + US	251	AUROC 0.8200	Imbalance-induced probability distortion unaddressed in downstream decision rules.
Ma et al. [5]	2025	Multiphase CT + WSI	274	AUROC 0.8363	Pre-fusion probability scaling omitted; unverified rank vs log-odds gain.
Zang et al. [2]	2025	Clinical + CT + WSI	1,648	C-index 0.8860	Multi-center cohort, but calibration–fusion hazard left unexamined.
Xiao et al. [13]	2026	3D CT + Pathology + Omics	Multi-center	AUROC 0.8400	Product-of-experts architecture without mandatory Platt pre-calibration.
RenoFusion (Ours)	**2026**	**Clinical + Genomic + 3D CT + LLM RAG**	**126 Aligned + 150 RAG**	**AUROC 0.8756 / RAGAS 0.5688**	**First to mandate Platt pre-calibration across both 3D decision fusion and RAG; achieves state-of-the-art calibrated AUROC 0.8756 (ECE = 0.0564); zero data leakage; out-of-fold 5-CV; LLM agentic RAG validation.**

This empirical finding aligns with emerging literature on tool-calling dynamics in domain-specific medical LLMs [24]: multi-round reasoning loops introduce extraneous meta-cognitive text that dilutes answer conciseness and term density on focused guideline corpora, without yielding commensurate accuracy gains over direct multimodal retrieval. In small, highly curated guideline settings, direct hybrid retrieval with calibrated evidence fusion offers a superior accuracy-to-efficiency profile compared to iterative agentic loops.

5.4. Comparative Analysis with Existing ccRCC Benchmarks

To contextualize the methodological advance of RenoFusion within the broader multimodal machine learning literature in renal oncology, Table 8 provides a systematic comparative synthesis of our framework alongside seven landmark ccRCC predictive benchmarks published between 2021 and 2026 [1–5, 12, 13].

Prior state-of-the-art studies [1, 2, 4, 12, 13] have reported uncalibrated fusion metrics ranging between 0.800 and 0.840. By integrating 3D volumetric deep representations (MedicalNet + PyRadiomics), hallmark pathway genomics, and clinical TNM staging, RenoFusion establishes a new state-of-the-art benchmark of **AUROC 0.8756 [0.7991–0.9392]** (AUPRC 0.4989, F_2 0.7143) while strictly maintaining probabilistic calibration fidelity (ECE = 0.0564).

RenoFusion advances beyond existing literature in five key dimensions:

1. **High-Performance 3D Volumetric Integration:** By leveraging 3D MedicalNet deep features and 3D shape/heterogeneity radiomics, RenoFusion elevates single-modality imaging discrimination to AUROC 0.8055, providing the necessary discriminative substrate for synergistic multimodal fusion (AUROC 0.8756).

2. **First Calibration-Aware Evaluation Protocol:** RenoFusion is the first multimodal framework in renal oncology to enforce mandatory per-fold out-of-fold Platt scaling prior to both decision fusion and retrieval-score fusion, guaranteeing that reported metrics reflect genuine clinical discrimination rather than probability-scale distortion.
3. **First LLM-Powered RAG Calibration Hazard Diagnosis:** RenoFusion provides the first empirical evidence that the Calibration–Fusion Hazard extends to retrieval-augmented generation, demonstrating catastrophic miscalibration (ECE = 0.5718) in uncalibrated rank fusion of heterogeneous retrieval streams.
4. **Zero-Leakage Rebalancing Guardrails:** Unlike prior pipelines that apply SMOTE globally or across validation splits, RenoFusion strictly isolates synthetic oversampling and feature selection to training folds, preventing probability tail leakage into validation sets.
5. **Real LLM Agentic RAG Validation:** Using Qwen2.5-3B-Instruct with ReAct tool-calling, RenoFusion validates that calibrated multimodal retrieval provides statistically significant improvement ($p = 0.001$) while agentic reasoning achieves balanced clinical correctness in domain-grounded clinical decision support.

The methodological warning extends beyond ccRCC. Maier-Hein et al. [20] cautioned that common calibration metrics can be highly model-dependent. Chen et al. [18] emphasized that learned risk scores must be subjected to post-hoc calibration regardless of the model's training-time probabilistic structure. Andersen et al. [19] demonstrated that SMOTE systematically degrades calibration across 10 clinical datasets despite preserving rank-based discrimination. The present study unites these observations under a single empirical focus: within a single multimodal ccRCC pipeline spanning both classifier fusion and LLM-powered RAG, the joint application of class rebalancing, information-theoretic fusion, and ad-hoc retrieval-score normalization can manufacture predictive claims that disappear under proper cross-calibration. Per-modality out-of-fold probability calibration before fusion is a mandatory precondition for valid performance reporting in both decision-level and retrieval-augmented clinical AI systems.

6. Study Limitations

This study is subject to several important methodological and clinical limitations:

1. **Small Multimodal Cohort and Feature Dimensionality:** The primary multimodal overlapping cohort contains $n = 126$ TCGA-KIRC patients with 18 M_1 metastatic cases (14.29% prevalence), constrained by the strict requirement for cross-modal alignment across clinical SEER records, RNA-seq genomic profiles, and contrast CT volumes [1, 25]. While leave-one-out sensitivity analysis showed modest AUROC variability ($|\Delta\text{AUROC}| \leq 0.0330$), fitting PCA-reduced 2,048-dimensional ResNet50 deep features on a small positive sample remains in the high-dimension/small-sample regime, and fusion-rule gain detection is inherently constrained by sample size.
2. **Base Model Asymmetry and Rebalancing Focus:** M1 Clinical operates near chance level (AUROC 0.5592) with high sensitivity (0.9444) reflecting operating-threshold choice rather than discriminative skill (precision 0.1717). Consequently, the absence of fusion gain partly reflects that one input stream is weak while M2 Genomic was SMOTE-inflated. Furthermore, while SMOTE was rigorously investigated and identified as contributing a +0.0438 artifactual boost, other class-rebalancing techniques (focal loss, random under-sampling, class weighting) were not systematically benchmarked.
3. **Guideline-Level Retrieval vs. Patient-Level Prediction in RAG:** The Stage II RAG benchmark was conducted on a curated 10-document ccRCC guideline corpus with 150 clinical decision queries, evaluating whether the LLM retrieves appropriate clinical management guidelines rather than performing patient-case similarity matching across raw population registries. With 10 documents and multi-label relevance, top-1 retrieval saturated (MRR = 1.0000), meaning MRR ceased to discriminate retrieval quality while Recall@3 and NDCG@3 provided the true measurement.
4. **Residual Miscalibration in Fused Retrieval:** While Platt calibration reduced the Expected Calibration Error of multimodal hybrid retrieval from 0.5718 to 0.3735 (and Brier score from 0.5625 to 0.3741), ECE = 0.3735 remains substantial and far from the well-calibrated single-stream baseline (ECE = 0.0633). Post-hoc monotonic scaling only partially repairs the severe probability distortion induced by combining unbounded lexical scores with cosine similarities, demonstrating that score-level rank fusion introduces structural miscalibration that requires natively calibrated retrieval architectures.
5. **LLM Self-Judgment Protocol:** Generation Faithfulness was evaluated via an LLM-as-Judge protocol using the same model backbone (Qwen2.5-3B-Instruct). While prompt-grounded token verification was enforced and confirmed zero factual fabrications against provided guideline passages, self-evaluation introduces a potential conflict of interest. Future studies must incorporate independent multi-model entailment verifiers (e.g., DeBERTa-v3 NLI) and formal blinded urologist review panels.
6. **Absence of External Multi-Center and Temporal Validation:** The entire pipeline was evaluated under 5-fold cross-validation without non-TCGA external cohorts or temporal validation (e.g., training on earlier registry years and testing on prospective years). Cross-institutional differences in CT acquisition parameters

(scanner vendor, slice thickness, reconstruction kernel) may introduce covariate shift not captured by internal cross-validation.

7. Future Work

The findings of RenoFusion establish a clear roadmap for subsequent research in multimodal clinical AI and oncology RAG:

1. **Population-Scale Patient-Case Retrieval RAG:** Expanding the retrieval corpus from 10 guideline documents to population-scale patient registries (such as the 44,158 SEER ccRCC clinical vignettes) to enable direct patient-case retrieval, where the retrieval target is true patient metastatic outcome at diagnosis under extreme class imbalance (6.41% prevalence).
2. **Class-Aware Corrective RAG (CRAG) and Memory Augmentation:** Developing natively calibrated, class-aware Corrective RAG architectures that dynamically evaluate retrieval confidence and trigger fallback search mechanisms when fused retrieval confidence falls below clinical reliability thresholds [26, 27].
3. **Temporal Registry Validation:** Implementing temporal split validation across SEER registry years (e.g., training on historical cases 2010–2017 and testing on prospective cases 2018–2020) to evaluate whether calibration stability persists under prospective clinical drift.
4. **Independent Multi-Model Verifiers and Clinical Expert Panels:** Replacing self-judged faithfulness with independent natural language inference (NLI) cross-encoders and blinded expert panel review by practicing urological oncologists to establish clinical safety bounds.
5. **Systematic Rebalancing Ablations:** Benchmarking focal loss, Class-Balanced Loss, and Random Under-Sampling against SMOTE to determine whether alternative rebalancing techniques mitigate log-odds stretching in decision-level fusion.

8. Conclusion

This study performed a zero-leakage multimodal evaluation of distant metastasis prediction in clear cell renal cell carcinoma across two complementary paradigms: decision-level classifier fusion and LLM-powered retrieval-augmented generation.

Stage I (Decision-Level Fusion): Benchmarked across three single-modality base models and seven decision-level fusion strategies under three calibration regimes on $n = 126$ TCGA-KIRC patients, single-modality CT radiomics achieved the highest hold-out discrimination (AUROC 0.7037, Platt BSS +0.0515). Uncalibrated information-theoretic fusion rules manufactured artifactual AUROC gains up to 0.7505 that dissolved under 5-fold Platt cross-calibration (Platt F_2-weighted AUROC 0.6631, $p = 0.779$; Platt Dempster–Shafer AUROC 0.6831, $p = 0.442$).

Stage II (LLM-Powered RAG): Evaluated across Naive, Multimodal Hybrid, and Agentic RAG architectures on 150 ccRCC clinical decision queries, Calibrated Multimodal RAG achieved statistically significant superiority over Naive RAG (ΔRAGAS $= +0.0481$, $p = 0.001$), with balanced class-stratified accuracy (M_1 Sensitivity = 0.7778, M_0 Specificity = 0.7685). Uncalibrated Reciprocal Rank Fusion produced catastrophic probability distortion (ECE = 0.5718), which Platt calibration partially mitigated (ECE = 0.3735). Agentic RAG exhibited an efficiency tradeoff, with multi-round reasoning overhead diluting answer relevance (RAGAS 0.4870 vs. 0.5688).

The central contribution of this work is the formal diagnosis and mechanistic decomposition of the **Retrieval–Fusion Calibration Hazard**: a unified evaluation artifact affecting both classifier ensembles and RAG systems that combine heterogeneous, uncalibrated scores. The Hazard originates from inter-modality error correlation ($\rho \in [0.3842, 0.4619]$), high evidence conflict ($\bar{K} = 0.4381$), SMOTE probability stretching (+0.0438 boost), and multi-source rank-mixing distortion (ECE = 0.5718). Per-modality out-of-fold probability calibration before fusion is a mandatory precondition for valid performance reporting across all multimodal biomedical AI systems.

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Declaration of Generative AI in Scientific Writing

During the preparation of this work, the authors used generative AI tools solely for linguistic refinement, grammar checking, and structural formatting of the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. The entire study design, experimental pipeline, code implementation, data curation, statistical evaluations, and scientific interpretations were independently developed and empirically verified by the authors through original computational experiments.

Data Availability Statement

The population-scale clinical dataset from the Surveillance, Epidemiology, and End Results (SEER) program contains confidential patient health information and is available upon request following execution of a formal Data Use Agreement with the National Cancer Institute (NCI) SEER program. The genomic RNA-sequencing data and clinical records for the TCGA-KIRC cohort are publicly available from the NCI Genomic Data Commons (GDC; `https://portal.gdc.cancer.gov/`). The 3D CT radiomics imaging volumes for the TCIA-KIRC and KiTS23 cohorts are publicly accessible from The Cancer Imaging Archive (TCIA; `https:`

//www.cancerimagingarchive.net/) and the KiTS23 Challenge repository (https://kits-challenge.org/kits23/). Complete source code, cross-validation evaluation pipelines, tabular metrics, and figure generation scripts for the RenoFusion framework are open-source and publicly available on GitHub at https://github.com/ali-Hamza817/RenoFusion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Authorship Contribution Statement

Ali Hamza: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization, Project administration.
Imran Usman: Conceptualization, Supervision, Methodology, Resources, Writing – Review & Editing, Project administration.

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